When Validation Changes the Answer: GARCH(1,1) versus Gradient Boosting for Forecasting S&P 500 ETF Volatility

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https://github.com/mateeniqbal3/vol-forecast-model-comparison

Abstract

This study asks whether a gradient-boosted tree model (LightGBM) forecasts the 5-day realized variance of SPY better than GARCH(1,1) and a RiskMetrics EWMA. It also asks whether the answer depends on how the models are validated. All three models are scored under a naive random split of forecast dates and under expanding-window walk-forward validation (26 annual folds, 2000–2025, 6,534 forecast dates). The walk-forward scheme purges training targets that overlap the test period. The ML design, the walk-forward protocol and the significance test were each fixed before the results that could have influenced them.

Under walk-forward validation, LightGBM's QLIKE advantage over GARCH(1,1) is 0.0124, which is not significant: Diebold–Mariano $p = 0.47$, 95% CI -0.046 to $+0.022$. Its RMSE is higher (1.589e-3 vs. 1.443e-3; $p = 0.18$). Under the naive split the same model appears to beat GARCH decisively on both losses (QLIKE $p < 0.0001$; MSE $p = 0.016$). On identical test dates the naive split overstates LightGBM's QLIKE by 0.065 and GARCH's by 0.008; for EWMA, which estimates nothing, it changes nothing.

The mechanism is identified directly. A tree ensemble's forecast is capped by the targets it was trained on. The model trained through 2007 cannot forecast above 0.0022 (about 33% annualized volatility) for any input, while the autumn of 2008 realized up to 0.035 (about 133%). That one year moves the full-sample QLIKE difference by +0.027, more than twice LightGBM's net advantage. The added complexity was not justified, and naive validation would have concluded the opposite.

Keywords: volatility forecasting, GARCH, gradient boosting, LightGBM, walk-forward validation, data leakage, QLIKE, Diebold–Mariano test, pre-registration

JEL classification: C52, C53, C58, G17

1. Introduction

Machine-learning models are routinely proposed as improvements on parametric volatility models. The comparison is easy to get wrong in two ways. The first is validation: volatility is persistent, and forecast targets built from overlapping windows are serially dependent. A random train/test split therefore places training observations immediately next to, and partly inside, the test observations. The second is selection: when many designs are tried and the best one is reported, even a correct validation scheme overstates out-of-sample performance.

This study compares a simple and a complex volatility forecaster under controls for both problems, and reports the result together with the answer a naive analysis would have given. Its contributions are modest and specific:

1. **A controlled contrast of validation schemes.** The naive and walk-forward forecasts of each model are scored on the same 1,321 dates. This isolates the effect of the validation scheme from the effect of the sample period. The naive split's bias grows with how much a model learns from individual target rows: 0 for EWMA, 0.008 QLIKE for GARCH(1,1), 0.065 for LightGBM.
2. **A pre-registered comparison.** The design of the complex model, the walk-forward protocol and the significance test with its decision rule were each written down and committed before the results they could have influenced. Exactly one ML configuration was built and evaluated.
3. **A directly measured mechanism.** The walk-forward failure of the tree model is traced to a forecast ceiling. Scaling the model's variance inputs by 100 leaves its forecast unchanged.

The finding echoes Hansen and Lunde (2005), who found little that outperforms GARCH(1,1) for exchange-rate volatility. It does so for a modern ML competitor, and with an explanation of *why* the competitor falls short.

2. Related work

Parametric volatility models. GARCH (Engle, 1982; Bollerslev, 1986) and the exponentially weighted moving average of RiskMetrics (J.P. Morgan and Reuters, 1996) are the standard benchmarks. Andersen and Bollerslev (1998) showed that the apparently poor fit of GARCH forecasts largely reflects the noise in squared-return proxies, not a failure of the model. Hansen and Lunde (2005) compared 330 ARCH-type models and found little evidence that any beats GARCH(1,1) for exchange-rate data. The HAR model of Corsi (2009) motivates the multi-horizon realized-variance features used here.

Evaluation with a noisy proxy. Patton (2011) showed that when forecasts are compared against a conditionally unbiased but noisy variance proxy, only certain losses rank them consistently. MSE and QLIKE on the variance scale are among them; RMSE on the volatility scale is not. This study therefore evaluates everything in variance units. Diebold and Mariano (1995) provide the test of equal predictive accuracy, with heteroskedasticity- and autocorrelation-consistent variance estimation (Newey and West, 1987, 1994).

Machine learning and financial validation. Gradient boosting (Friedman, 2001), in the LightGBM implementation (Ke et al., 2017), is a common default for tabular prediction. López de Prado (2018) stresses that cross-validation on financial data leaks information through overlapping labels and must be purged. Recent studies apply ML to volatility forecasting with intraday data (e.g. Christensen, Siggaard and Veliyev, 2023). This study works at the daily frequency that free data allow, and puts its emphasis on validation and pre-registration rather than on model breadth.

3. Data and forecasting target

Data.

- **Source:** daily split- and dividend-adjusted OHLCV for the SPDR S&P 500 ETF (SPY), from `yfinance`.
- **Sample:** its first trading day (1993-01-29) to 2025-12-31, giving 8,287 daily log returns.
- **Cleaning:** no rows had duplicate dates or missing closes, so none were dropped.
- **Drift check:** a data manifest records checksums of the return series, so a re-download can be checked for provider revisions.

SPY was chosen over the `^GSPC` index for its genuine traded volume. It was chosen over single stocks because index volatility is less dominated by company-specific events.

Stylized facts. The full-sample returns show the familiar features of equity-index returns (exploratory analysis, `notebooks/01_eda.ipynb`):

- annualized volatility of 18.6%;
- excess kurtosis of 11.4 and skewness of -0.25;
- strong volatility clustering: the Ljung–Box statistic at lag 20 is about 12,100 for |r| against 126 for r;
- persistent weekly volatility: on non-overlapping weeks the autocorrelation of log RV is 0.58 at one week and 0.11 at 52 weeks;
- a leverage effect: the rank correlation between r_t and the log of the next week's realized variance is -0.135.

No model choice was made on the basis of this analysis.

Target. A forecast is made at the close of day t for the realized variance of the next five trading days,

$$RV_t = \sum_{k=1}^{5} r_{t+k}^2,$$

where $r_t = \ln(P_t/P_{t-1})$. The target uses only returns strictly after t. It is fully observed at the close of t+5, which governs the purge in Section 5.

Five-day RV has a mean of 6.9e-4 and a median of 3.0e-4. It is heavily right-skewed in levels and close to symmetric in logs. No week has RV = 0.

4. Models

All three models produce a forecast F_t of RV_t in variance units.

EWMA. The model is

$$\hat{\sigma}^2_{t+1|t} = \lambda \hat{\sigma}^2_{t|t-1} + (1-\lambda) r_t^2,$$

with $\lambda = 0.94$ fixed at its RiskMetrics daily value. Variance is treated as a random walk, so $F_t = 5\,\hat{\sigma}^2_{t+1|t}$. Nothing is estimated.

GARCH(1,1). The model is

$$\sigma^2_{t+1|t} = \omega + \alpha r_t^2 + \beta \sigma^2_{t|t-1},$$

with zero mean, estimated by Gaussian quasi-maximum likelihood, which is consistent under fat tails (Bollerslev and Wooldridge, 1992). The forecast sums the analytic 1- to 5-step forecasts,

$$\sigma^2_{t+k|t} = \bar{v} + (\alpha+\beta)^{k-1}\left(\sigma^2_{t+1|t} - \bar{v}\right), \qquad \bar{v} = \omega/(1-\alpha-\beta).$$

Estimation and filtering are separate. Parameters are estimated on the training information set. The recursion then runs forward with the parameters fixed, starting from $\bar{v}$, so every forecast uses only data up to *t*.

The Gaussian one-step likelihood is itself a QLIKE loss. The estimator and the primary evaluation metric therefore share one criterion. GJR-GARCH (Glosten, Jagannathan and Runkle, 1993) was deliberately not used. Choosing the baseline because the full-sample analysis showed a leverage effect would have been a data-driven specification choice.

LightGBM. The ML model is a gradient-boosted regression tree ensemble trained with the gamma objective and a log link. The gamma deviance of a forecast *F* is

$$2\left(\frac{RV}{F} - \ln\frac{RV}{F} - 1\right),$$

which equals the QLIKE loss $\ln F + RV/F$ up to terms that do not depend on *F*. The model is therefore trained on the primary evaluation metric, and it predicts the conditional mean of RV directly, avoiding the bias of back-transforming a log-scale regression.

It uses 18 features, all computed at the close of *t* from SPY's own OHLCV, with at most 252 days of history:

- mean squared returns over 1, 5, 22, 66 and 252 days (HAR-style);
- summed returns over 1, 5 and 22 days (signed, to capture leverage);
- downside semivariance over 5 and 22 days (Barndorff-Nielsen, Kinnebrock and Shephard, 2010; Patton and Sheppard, 2015);
- the Parkinson (1980) high–low range estimator over 1, 5 and 22 days;
- the squared overnight gap over 1 and 5 days;
- two relative-volume measures;

- the log drawdown from the 252-day high.

Other tickers such as VIX were excluded, because they change the information set rather than the model's complexity. Baseline forecasts were excluded as inputs, because they would turn the model into a stack of the baseline rather than an alternative to it.

Hyperparameters. They are chosen inside each fit, from the training dates alone:

1. The training dates are split in time order into an inner-training set (first 80%) and an inner-validation set (last 20%). Validation dates within five rows of the last inner-training date are purged.
2. For each of 12 grid points, `num_leaves` $\in \{7, 15, 31\}$ × `min_child_samples` $\in \{50, 200\}$ × `reg_lambda` $\in \{0, 10\}$, the model is trained with learning rate 0.03, row and column subsampling at 0.8 and early stopping (100 rounds, at most 2,000 trees).
3. The grid point with the lowest inner-validation QLIKE is refitted on all training dates with its early-stopped number of trees.

Across the 26 walk-forward folds this gives between 36 and 412 trees.

5. Validation design

The information set. Each model sees only the data its training rows touch. A training row at t has features dated up to t and a target built from returns t+1 to t+5. The data passed to the model therefore end five rows after the last training date. The same rule governs both schemes below, and tests enforce it.

Eligible dates. Every date with a complete target is eligible, after a one-year burn-in shared by all models. That gives 8,030 dates, from 1994-01-28 to 2025-12-23.

Naive random split (contrast only).

- The eligible dates are shuffled once with a fixed seed and split 80/20: 6,424 training and 1,606 test dates, interleaved in time.
- Almost every test date has training dates beside it, whose targets share up to four of its five returns.
- For GARCH, the information-set rule means estimation on the full sample. With a random split, training rows touch almost every return, so that is what the split amounts to, not an extra choice.

Walk-forward validation (the reported result).

- **Folds:** an expanding window with one test block per calendar year, 2000–2025 (26 folds).
- **Purge:** the fold whose first test date is T_0 trains on eligible dates at least five rows before T_0. Every training target is then observed by the close of T_0, and the data passed to the model end at exactly T_0.

- **Refits:** each model is refitted once per fold and forecasts every date of that year with its parameters fixed. Its inputs still update daily.
- **Size:** the first fold trains on 1,496 dates, and the test set has 6,534 dates.
- **Tests:** they verify that no test date precedes or overlaps its training window, that unobserved targets are purged, and that each fold's data end at its first test date.

Order of work. Each design was committed before the results that could have influenced it:

Step	Fixed	Before any
1	Baseline specifications; naive-split protocol; baseline naive scores recorded	ML model existed
2	ML design, including features, objective, grid and training procedure	ML score, and any walk-forward score for any model
3	Walk-forward design	walk-forward score
4	Diebold–Mariano design and verdict rule	p-value (the step-3 point estimates had been seen)

The configuration log contains one entry. No feature set, objective, grid or procedure other than those described above was built or scored.

6. Evaluation

Losses. Both losses are on the variance scale: the primary metric QLIKE, $L = \ln F_t + RV_t/F_t$, and MSE, reported as RMSE. Lower is better for both.

Test. Differences in expected loss are tested with the Diebold–Mariano statistic on $d_t = L_t^A - L_t^B$. The variance is Newey–West (Bartlett) with lag $\max(4, \lfloor 4(n/100)^{2/9} \rfloor)$, which is 10 for the walk-forward sample and 7 for the naive one. A sensitivity lag of 63, about one quarter, allows for loss differences that stay correlated through volatility regimes. p-values are two-sided, normal.

Decision rule. It was fixed before any p-value was computed. The primary comparison is LightGBM against GARCH(1,1) on walk-forward QLIKE:

- *Justified:* mean $d_t < 0$ with $p < 0.05$ at both lags.
- *Weak evidence:* $p < 0.05$ at the primary lag only.
- *Not justified:* anything else.

GARCH(1,1) was named as the comparator because it is the estimated simple model. MSE is reported alongside QLIKE, and any disagreement between them is stated rather than resolved by choosing one.

7. Results

7.1 The comparison

Table 1 reports each model's losses under both schemes. The middle columns score the same 1,321 dates: the naive test dates from 2000 on, all of which also have a walk-forward forecast.

Table 1. Mean QLIKE and RMSE (lower is better)

Model	Naive, all (n = 1,606)	Naive, same dates (n = 1,321)	Walk-forward, same dates (n = 1,321)	Walk-forward, all (n = 6,534)
QLIKE				
EWMA	−6.6257	−6.5997	−6.5997	−6.5887
GARCH(1,1)	−6.6671	−6.6438	−6.6356	−6.6327
LightGBM	−6.7413	−6.7240	−6.6586	−6.6451
RMSE				
EWMA	1.490e-03	1.607e-03	1.607e-03	1.508e-03
GARCH(1,1)	1.403e-03	1.508e-03	1.545e-03	1.443e-03
LightGBM	1.328e-03	1.426e-03	1.680e-03	1.589e-03

Table 2. Diebold–Mariano tests. The difference is the first model's mean loss minus the second's; negative values favor the first model.

Scheme	Metric	Pair	Mean difference	DM (lag)	p	p at lag 63
Walk-forward	QLIKE	LightGBM − GARCH	−0.0124	−0.71 (10)	**0.47**	0.65
Walk-forward	QLIKE	LightGBM − EWMA	−0.0565	−2.56 (10)	0.011	0.070
Walk-forward	QLIKE	GARCH − EWMA	−0.0440	−3.87 (10)	0.0001	0.0002
Walk-forward	MSE	LightGBM − GARCH	+4.45e-07	+1.34 (10)	0.18	0.36
Walk-forward	MSE	LightGBM − EWMA	+2.54e-07	+0.74 (10)	0.46	0.60
Walk-forward	MSE	GARCH − EWMA	−1.92e-07	−2.10 (10)	0.036	0.12
Naive	QLIKE	LightGBM − GARCH	−0.0742	−6.08 (7)	< 0.0001	< 0.0001
Naive	MSE	LightGBM − GARCH	−2.06e-07	−2.42 (7)	0.016	0.018

Walk-forward. Under walk-forward validation, LightGBM's QLIKE advantage over GARCH(1,1) is 0.0124, with a 95% confidence interval of −0.046 to +0.022, far from significance. On MSE it is worse, though not significantly. GARCH(1,1) beats EWMA on QLIKE robustly at both lags. LightGBM beats EWMA at the primary lag but not at lag 63.

Naive split. Under the naive split, LightGBM beats GARCH on both losses. The QLIKE difference is six times larger than under walk-forward, and it is significant at either lag.

7.2 What the naive split changes

Scoring each model's naive and walk-forward forecasts on the same dates isolates the scheme effect (Table 3).

Table 3. Naive minus walk-forward loss on the 1,321 common dates

Model	QLIKE difference	p	MSE difference	p
EWMA	0 (identical forecasts)	—	0	—
GARCH(1,1)	−0.0082	0.033	−1.15e-07	0.094
LightGBM	−0.0654	0.004	−7.87e-07	0.14

The size of the bias follows how the model learns.

- **EWMA** estimates nothing, so the scheme cannot matter.
- **GARCH** estimates three parameters from a return series. The naive split gives it the full sample, including the test weeks. That is a small advantage, because the parameters move slowly: across the walk-forward folds α rises from 0.06 to 0.11 and β falls from 0.94 to 0.88. The full-sample estimates are $\alpha = 0.111$ and $\beta = 0.873$.
- **LightGBM** is fitted row by row to targets, and the naive split flatters it eight times as much as GARCH.

Two paths contribute.

- **Crisis weeks enter training.** On the 115 naive test dates in 2008 and 2020, realized variance averaged 0.00284. The naive-split LightGBM forecast 0.00185 on average; the walk-forward LightGBM forecast 0.00127. GARCH forecast 0.00232 and 0.00233 under the two schemes. Of LightGBM's −0.065 scheme effect, 2008 contributes −0.027.
- **Leakage from neighboring targets.** Outside 2008 and 2020 the naive split still flatters LightGBM by 0.038, against 0.007 for GARCH. This is consistent with the model learning from adjacent training targets that share four of five returns with each test target. Slow-moving features such as the 252-day variance, drawdown and relative volume make it easy for trees to localize a date in time. The two paths were not measured separately. The split by year was made after seeing the results and is descriptive.

The effect on the conclusion. On QLIKE the naive split keeps the same point-estimate winner but inflates LightGBM's lead. On the common dates the lead is 0.080 under the naive split and 0.023 under walk-forward. On RMSE the ranking reverses: LightGBM is best under the naive split and worst of the three models under walk-forward.

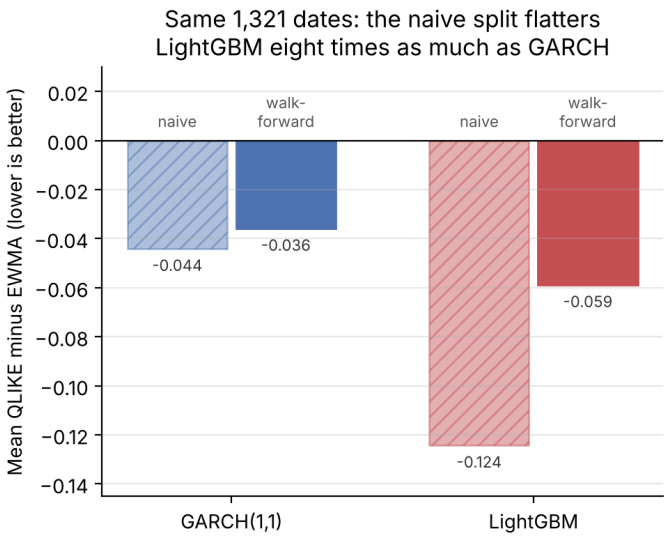


Figure 1. QLIKE improvement over EWMA on the 1,321 common dates, under the naive split and under walk-forward validation.

7.3 Mechanism: the forecast ceiling of a tree ensemble

A boosted tree ensemble predicts by summing leaf values, and each leaf value summarizes the training targets that reach it. Inputs beyond the most extreme training inputs fall into the same boundary leaves, so the forecast stops rising.

The 2008 fold. This model was trained on 1994 to late 2007. The selected grid point was 31 leaves, at least 50 training dates per leaf and L2 penalty 10, with 83 trees.

- The largest training target was 0.0099, about 70% annualized volatility.
- In the autumn of 2008 the model's inputs left that range. The 22-day realized variance reached 4.2 times its training maximum and was above it on 65 test dates. The 5-day Parkinson estimator reached 3.1 times its training maximum.
- Realized 5-day variance exceeded the largest training target on 26 test dates. It peaked at 0.0352 (about 133% annualized) for the forecast made on 2008-10-08.
- On that date the model forecast 0.00216. Multiplying all eleven variance-type inputs by 2, 10 or 100 leaves the forecast at exactly 0.00216.
- The largest forecast this model makes on any date in the sample is 0.00222, about 33% annualized. That is well below even its own largest training target, because each leaf averages many training dates.

GARCH(1,1) has no such bound. Its forecast is linear in r_t^2, and it reached 0.0138 (about 83% annualized) in the same weeks. That is still far below realized variance, but six times the tree model's ceiling.

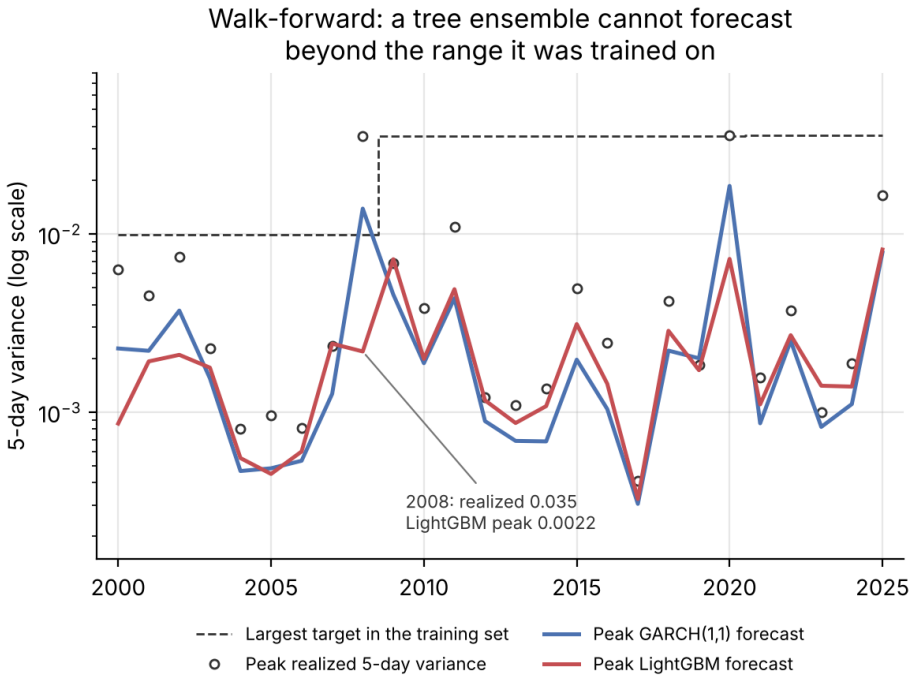


Figure 2. Per walk-forward fold: the largest training target, peak realized variance, and each model's peak forecast.

Consequence for the comparison. In 2008 LightGBM's QLIKE is worse than GARCH's by 0.70 per date. Averaged over all 6,534 dates, that one year adds +0.027 to the mean difference, more than twice LightGBM's net advantage of 0.012 (Appendix Table A1).

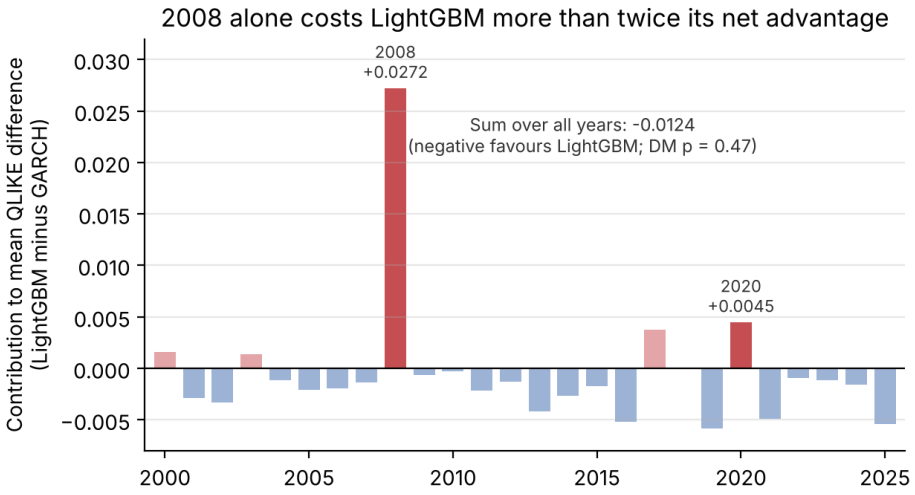


Figure 3. Each year's contribution to the full-sample walk-forward QLIKE difference, LightGBM minus GARCH(1,1).

Other volatile years. A consistent pattern appears across them. It was examined after the results

were known and is descriptive:

- *Before 2008 was in training (the 2000–2008 folds):* LightGBM's largest forecast in any test year never exceeded 0.0024.
- *After 2008 entered training:* its peak forecasts reached 0.0072 (2009 and 2020) and 0.0081 (2025).
- *Shocks within that range:* in 2011 (realized peak 0.0109) and 2025 (0.0164), the peak LightGBM forecast matched or exceeded GARCH's, and LightGBM had the lower loss in both years.
- *A shock of 2008's size:* in 2020 (realized peak 0.0357), LightGBM's peak forecast was 0.0072 against GARCH's 0.0185, and it lost the year.

7.4 Why QLIKE and RMSE disagree

Under walk-forward, LightGBM has the best QLIKE and the worst RMSE. The two losses weight different weeks.

MSE is decided by the extremes. The top 1% of weeks by realized variance (66 dates) account for 83% of LightGBM's total squared error, 68% of GARCH's and 65% of EWMA's. LightGBM's mean forecast is 0.79 times mean realized variance; GARCH's and EWMA's are 1.00. The shortfall sits in the extreme weeks, where the ceiling binds.

QLIKE scores relative errors. Each week contributes on a comparable scale, so the typical week dominates. On the remaining 99% of dates LightGBM's QLIKE is -6.724, against -6.686 for GARCH. Its RMSE there is also the lowest, 6.5e-4 against 8.2e-4. This split is descriptive, not a test.

LightGBM is worse in the extreme weeks on both losses. Its QLIKE in the top 66 weeks is $+1.05$, against -1.38 for GARCH. But those weeks are 1% of the sample, and QLIKE averages over all weeks.

Year by year, LightGBM has the lower QLIKE in 21 of 26 years and the lower MSE in 18. It is worse on QLIKE in 2000, 2003, 2008, 2017 and 2020.

8. Discussion

The verdict. By the pre-registered rule, the added complexity is *not justified*. LightGBM's walk-forward QLIKE advantage over GARCH(1,1) is not statistically distinguishable from zero, and its RMSE is higher. This is an absence of evidence for improvement, not evidence of equivalence. The confidence interval admits a LightGBM advantage of up to about 0.046 QLIKE.

When is complexity worth it? The two steps up in complexity behaved differently. Moving from EWMA to GARCH(1,1) adds three estimated parameters. It buys a QLIKE improvement of 0.044 that holds at both lags. That is consistent with mean reversion and an estimated reaction to shocks describing index volatility well. Moving from GARCH to a flexible nonparametric model buys a small, fairly consistent gain in ordinary conditions. It gives that gain back when volatility leaves the

range the model has seen, which is when a volatility forecast matters most. GARCH's functional form extrapolates by construction; a tree ensemble's does not.

Why naive validation is especially misleading for flexible models. The naive split's bias scales with how much a model learns from individual target rows. For a flexible model it hides the failure mode that decides the out-of-sample comparison: a random split gives the model training examples from the very crises it is later tested on.

What might change the result. Two kinds of design could remove the ceiling: trees with linear leaves, or a model of RV relative to a GARCH forecast rather than of RV itself. Either would let a flexible model extrapolate. Neither was tried here, because doing so after seeing these results would be the post-hoc redesign the study's protocol rules out. They are hypotheses for a separate, pre-registered study.

9. Limitations

- **Scope.** The study covers one instrument, daily data, a 5-day horizon and 2000–2025 out of sample. The results say nothing directly about single stocks, other asset classes or intraday horizons.
- **Proxy noise and power.** A sum of five squared daily returns is a noisy proxy for integrated variance. An intraday realized-variance target would rank the same forecasts more sharply and could narrow the confidence interval enough to be decisive.
- **Two episodes dominate.** The net walk-forward comparison rests heavily on 2008 and 2020. The figures that exclude those years are descriptive, chosen after seeing the results, and are not evidence about calm markets.
- **Information versus flexibility.** LightGBM saw the high–low range, the open and volume, which the baselines did not. It still showed no significant gain. No experiment separates the value of the extra inputs from the value of the model.
- **One protocol.** The study used one ML configuration, one walk-forward scheme (expanding window, annual refits) and a symmetric GARCH baseline. A rolling window, more frequent refits or an asymmetric baseline might change the magnitudes.
- **Naive-split tests.** The DM tests on the naive split treat the sorted random test dates as a time series. They are reported only as a contrast.
- **Data.** `yfinance` is a free, unofficial source, and adjusted history can be revised. The recorded checksums make any such drift detectable.

10. Conclusion

Under proper walk-forward validation, a gradient-boosted tree model with more information and a training loss matched to the evaluation metric did not forecast SPY's weekly variance detectably better than GARCH(1,1). A naive random split would have reported the opposite, with high confidence, on both standard losses. The mechanism is concrete and measurable. A tree ensemble

cannot forecast beyond the targets it was trained on. The naive split conceals this by training on the crises it tests. The lesson is narrower than "simple beats complex". A flexible model's out-of-sample value in volatility forecasting depends on whether it can extrapolate into regimes it has not seen, and only a validation scheme that withholds those regimes can tell.

Data and code availability

All code, tests, the design record (`DECISIONS.md`) and every recorded output are available at https://github.com/mateeniqbal3/vol-forecast-model-comparison under the MIT License, and each release is archived on Zenodo (the DOI is listed in the repository README). Raw prices are not redistributed; the data manifest records checksums of the return series used. Appendix B lists the command that regenerates each output. The figures are produced by `python paper/make_figures.py` from the committed result files only.

Acknowledgements and AI-use disclosure

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Appendix A. Walk-forward results by year

Table A1. Δ is LightGBM minus GARCH(1,1); negative values favor LightGBM. "Training max RV" is the largest target in that fold's training set. The last column counts test dates whose realized RV exceeded it. The yearly QLIKE differences, weighted by each year's share of dates, sum to the full-sample difference of −0.0124.

Year	n	QLIKE GARCH(1,1)	QLIKE LightGBM	Δ QLIKE	Δ MSE	Max realized RV	Training max RV	Dates above training max
2000	252	−5.7562	−5.7141	+0.0421	+6.82e-08	0.0063	0.0099	0
2001	248	−5.9895	−6.0661	−0.0765	−1.88e-07	0.0045	0.0099	0
2002	252	−5.6494	−5.7347	−0.0853	−2.28e-07	0.0074	0.0099	0
2003	252	−6.6716	−6.6361	+0.0355	+1.05e-08	0.0023	0.0099	0
2004	252	−7.2752	−7.3041	−0.0290	−4.83e-09	0.0008	0.0099	0
2005	252	−7.4344	−7.4884	−0.0540	−6.86e-09	0.0010	0.0099	0
2006	251	−7.5948	−7.6459	−0.0511	−7.14e-09	0.0008	0.0099	0
2007	251	−6.5671	−6.6035	−0.0363	+1.80e-08	0.0023	0.0099	0
2008	253	−5.2103	−4.5071	**+0.7033**	**+1.34e-05**	0.0352	0.0099	**26**
2009	252	−5.8550	−5.8724	−0.0173	+6.90e-08	0.0068	0.0352	0
2010	252	−6.4879	−6.4958	−0.0079	−2.99e-08	0.0038	0.0352	0
2011	252	−6.0699	−6.1255	−0.0556	−1.51e-07	0.0109	0.0352	0
2012	250	−6.9905	−7.0240	−0.0335	−1.65e-08	0.0012	0.0352	0
2013	252	−7.2833	−7.3911	−0.1078	−1.64e-08	0.0011	0.0352	0
2014	252	−7.2837	−7.3522	−0.0685	−9.81e-09	0.0014	0.0352	0
2015	252	−6.6611	−6.7068	−0.0457	−4.30e-08	0.0049	0.0352	0
2016	252	−7.0385	−7.1728	−0.1343	−2.65e-08	0.0025	0.0352	0
2017	251	−8.1022	−8.0050	+0.0972	+8.35e-09	0.0004	0.0352	0
2018	251	−6.5094	−6.5116	−0.0022	+7.82e-09	0.0042	0.0352	0
2019	252	−7.1429	−7.2951	−0.1521	−6.65e-08	0.0018	0.0352	0
2020	253	−5.6190	−5.5030	**+0.1160**	**+2.56e-07**	0.0357	0.0352	**3**
2021	252	−6.9494	−7.0769	−0.1275	−2.59e-08	0.0016	0.0357	0
2022	251	−5.7224	−5.7469	−0.0246	−1.02e-07	0.0037	0.0357	0
2023	250	−7.0077	−7.0379	−0.0302	−1.70e-09	0.0010	0.0357	0
2024	252	−7.0485	−7.0893	−0.0408	−1.39e-08	0.0019	0.0357	0
2025	245	−6.5385	−6.6831	−0.1446	−1.41e-06	0.0164	0.0357	0

Appendix B. Reproducibility

Every number in this paper is produced by the repository's pipeline and stored in version-controlled outputs. The exception is the stylized facts in Section 3, which are printed by the executed EDA notebook (`notebooks/01_eda.ipynb`).

Output	Command	Contents
`data/dataset_manifest.json`	`python -m src.data`	Sample, cleaning counts, return checksums
`docs/phase2_naive_baselines.json`	`python -m src.evaluate naive-baselines`	Baseline naive scores, recorded before the ML model existed
`docs/phase4_validation_results.json`	`python -m src.evaluate validate`	All models under both schemes; per-fold fitted parameters
`docs/phase5_comparison.json`	`python -m src.evaluate compare`	Tables 1–3, A1; DM tests; ceiling probe; error concentration

The outputs are deterministic. Rerunning `validate` fails if the baseline naive scores no longer match the record made before the ML model existed. The test suite (92 tests, synthetic data) covers look-ahead freedom of the target and of every feature, agreement of the GARCH filter with `arch`, the walk-forward split and purge, and the Newey–West variance against `statsmodels`. The design record, including the configuration log and the pre-registered decision rule, is in `DECISIONS.md`.